

# Sistemas Operativos

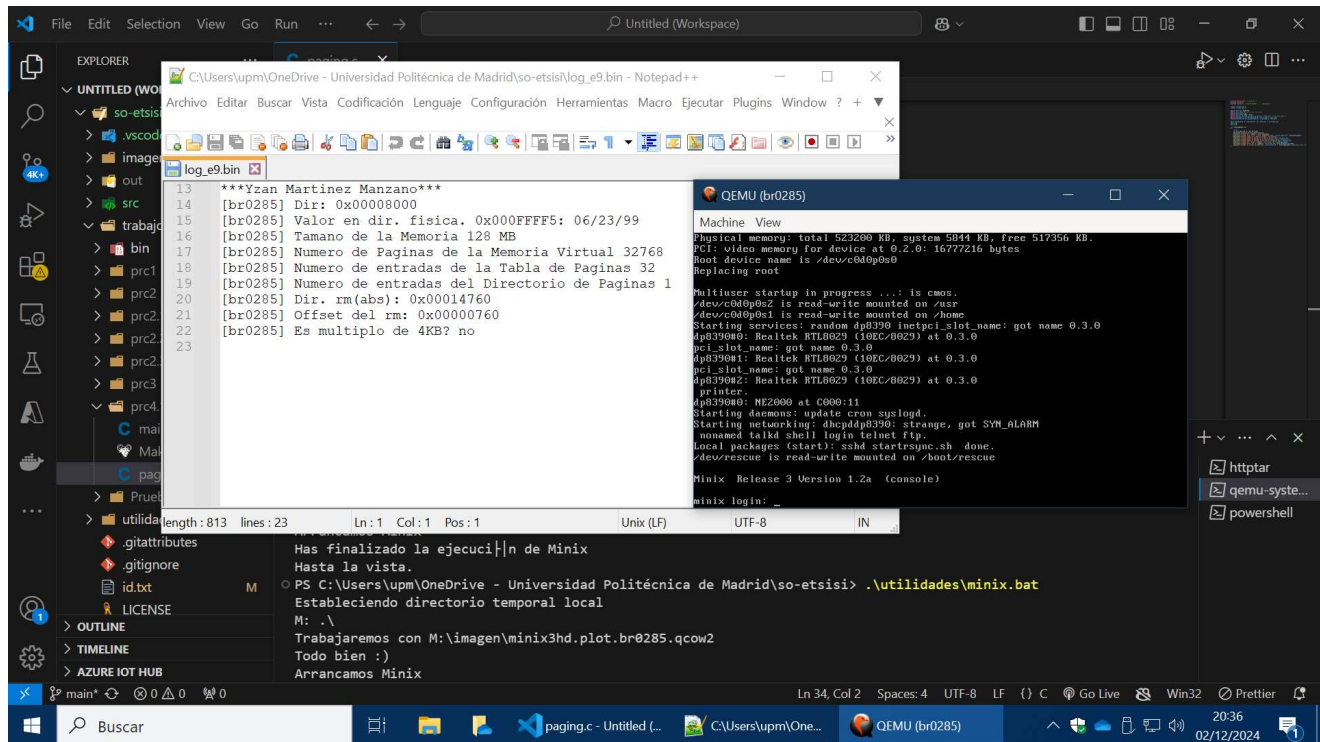
## Práctica 4.1

Fecha: 02/12/2024

Alumno: Yzan Martinez Manzano

Correo: [yzan.martinez.manzano@alumnos.upm.es](mailto:yzan.martinez.manzano@alumnos.upm.es)

## 1. Reserva de memoria



```
#include "e9_system.h" /* e9_printf */
#include "kernel.h"
#include <sys/vm.h> /* I386_VM_PRESENT, ... , I386_CR0_PG */

typedef unsigned pte_t;
typedef unsigned pde_t;

#define BIOS_DATE 0x00FFFF5
#define CEIL_DIV(x, y) (((x) + (y) - 1) / (y))
#define MEM_SIZE_MB 128
#define MB_TO_BYTE (1024 * 1024)
#define NUM_PAGS_VM (MEM_SIZE_MB * MB_TO_BYTE) / PAGE_SIZE /* Da 32768 bytes */
#define NUM_PAGS_TP CEIL_DIV(NUM_PAGS_VM * sizeof(pte_t), PAGE_SIZE)
#define NUM_PAGS_DP CEIL_DIV(NUM_PAGS_TP * sizeof(pde_t), PAGE_SIZE)

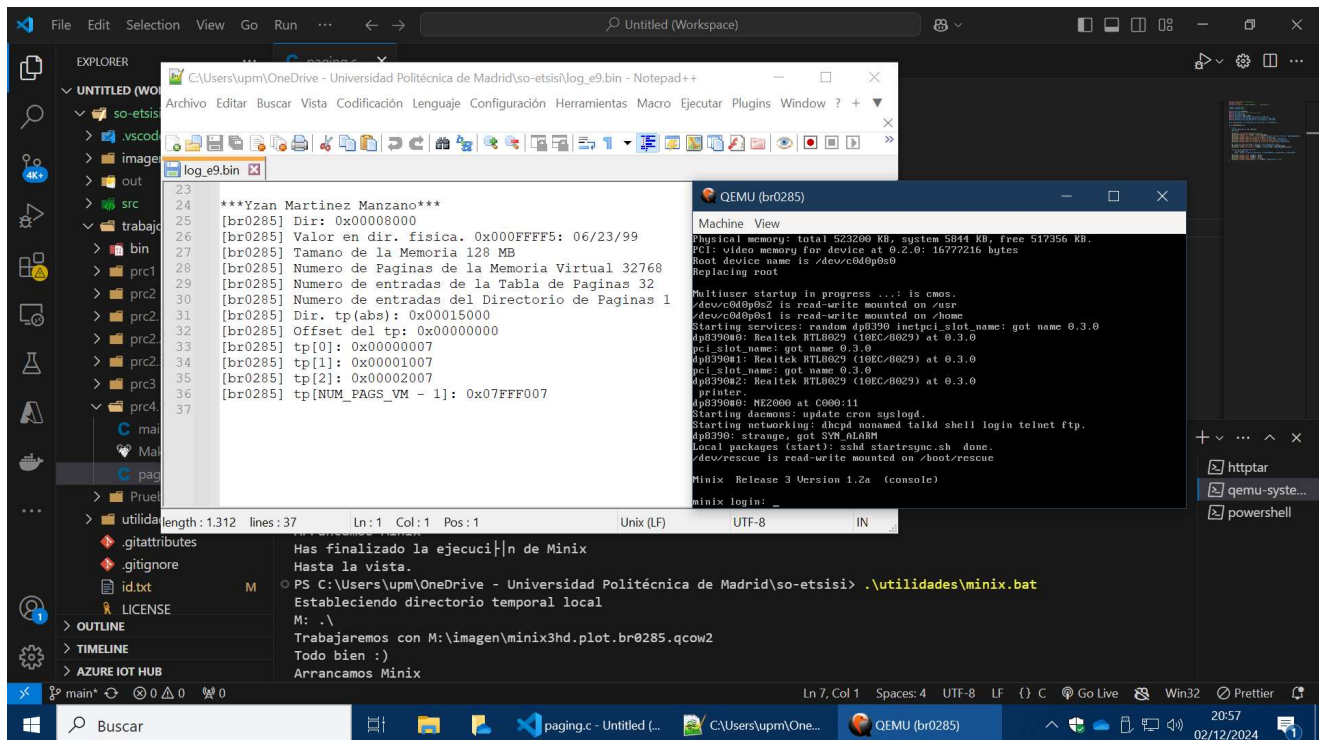
pde_t rm[(NUM_PAGS_TP + NUM_PAGS_DP + 1) * PAGE_SIZE / sizeof(pde_t)];
pde_t* tp;
pde_t* dp;
```

```

void setup_paging(void)
{
    int i;
    unsigned marco_tp, m, cr0, marco_dp;
    e9_printf("\n***Yzan Martinez Manzano***\n");
    e9_printf("[br0285] Dir: 0x%08X\n", kinfo.data_base);
    e9_printf("[br0285] Valor en dir. fisica. 0x000FFFF5: %s\n", (char*) (BIOS_DATE -
kinfo.data_base));
    e9_printf("[br0285] Tamano de la Memoria %u MB\n", MEM_SIZE_MB);
    e9_printf("[br0285] Numero de Paginas de la Memoria Virtual %u\n", NUM_PAGS_VM);
    e9_printf("[br0285] Numero de entradas de la Tabla de Paginas %u\n", NUM_PAGS_TP);
    e9_printf("[br0285] Numero de entradas del Directorio de Paginas %u\n", NUM_PAGS_DP);
    e9_printf("[br0285] Dir. rm(abs): 0x%08X\n", ((unsigned) rm) + kinfo.data_base);
    e9_printf("[br0285] Offset del rm: 0x%08X\n", ((unsigned) rm) + kinfo.data_base &
0xFFFF);
    e9_printf("[br0285] Es multiplo de 4KB? %s\n", (((unsigned) rm) + kinfo.data_base) %
PAGE_SIZE == 0 ? "si" : "no");
}

```

## 2. Tablas de páginas



```

#include "e9_system.h" /* e9_printf */
#include "kernel.h"
#include <sys/vm.h> /* I386_VM_PRESENT, ... , I386_CR0_PG */

typedef unsigned pte_t;
typedef unsigned pde_t;

#define BIOS_DATE 0x00FFFF5

```

```

#define CEIL_DIV(x, y) (((x) + (y) - 1) / (y))
#define MEM_SIZE_MB 128
#define MB_TO_BYTE (1024 * 1024)
#define NUM_PAGS_VM (MEM_SIZE_MB * MB_TO_BYTE) / PAGE_SIZE
#define NUM_PAGS_TP CEIL_DIV(NUM_PAGS_VM * sizeof(pte_t), PAGE_SIZE)
#define NUM_PAGS_DP CEIL_DIV(NUM_PAGS_TP * sizeof(pde_t), PAGE_SIZE)

pde_t rm[(NUM_PAGS_TP + NUM_PAGS_DP + 1) * PAGE_SIZE / sizeof(pde_t)];

void setup_paging(void)
{
    int i;
    unsigned marco_tp, m, cr0, marco_dp;
    pde_t* tp;

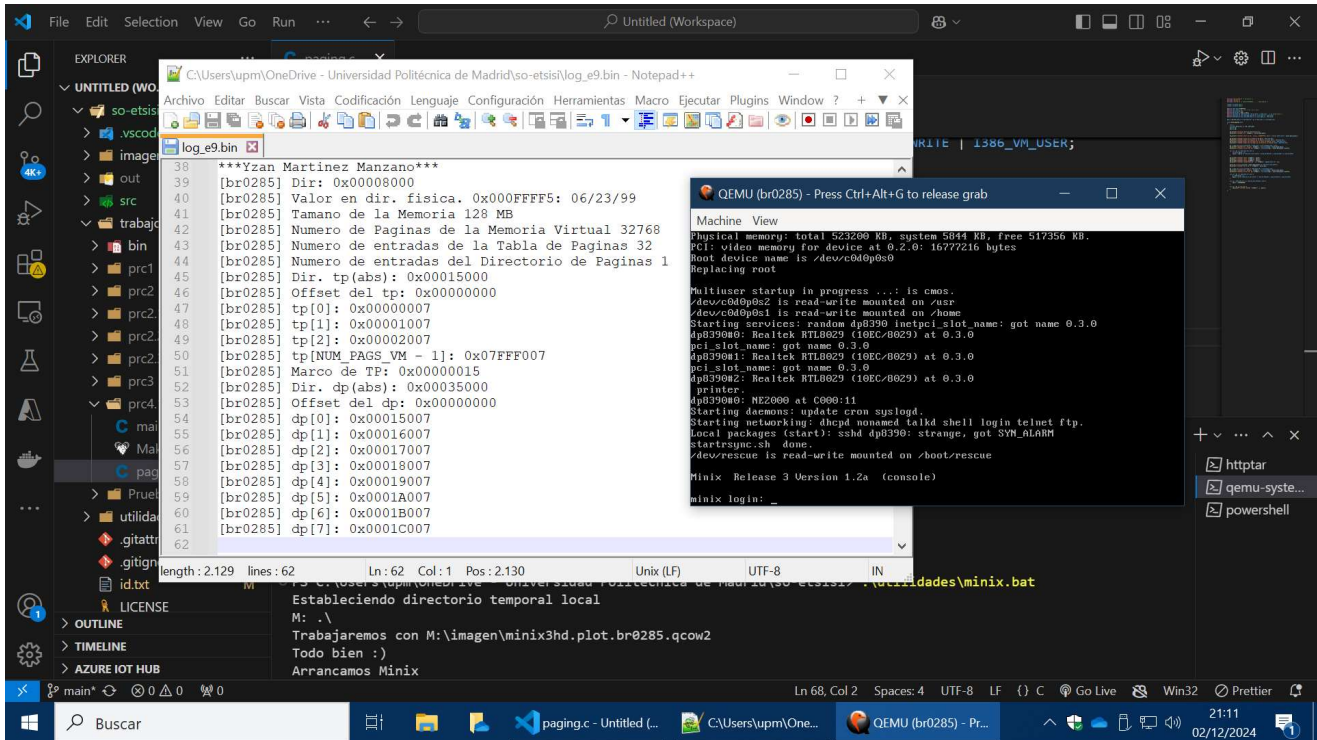
    e9_printf("\n***Yzan Martinez Manzano***\n");
    e9_printf("[br0285] Dir: 0x%08X\n", kinfo.data_base);
    e9_printf("[br0285] Valor en dir. fisica. 0x000FFFF5: %s\n", (char*) (BIOS_DATE -
kinfo.data_base));
    e9_printf("[br0285] Tamano de la Memoria %u MB\n", MEM_SIZE_MB);
    e9_printf("[br0285] Numero de Paginas de la Memoria Virtual %u\n", NUM_PAGS_VM);
    e9_printf("[br0285] Numero de entradas de la Tabla de Paginas %u\n", NUM_PAGS_TP);
    e9_printf("[br0285] Numero de entradas del Directorio de Paginas %u\n", NUM_PAGS_DP);

    tp = (pde_t*) (((unsigned)rm) + PAGE_SIZE - 1) & (~(PAGE_SIZE - 1));
    e9_printf("[br0285] Dir. tp(abs): 0x%08X\n", ((unsigned)tp) + kinfo.data_base);
    e9_printf("[br0285] Offset del tp: 0x%08X\n", ((unsigned)tp) + kinfo.data_base &
0xFFFF);

    for (i = 0; i < NUM_PAGS_VM; i++) {
        unsigned marco = i;
        tp[i] = (marco << I386_VM_PT_ENT_SHIFT) | I386_VM_PRESENT | I386_VM_WRITE |
I386_VM_USER;
    }
    e9_printf("[br0285] tp[0]: 0x%08X\n", tp[0]);
    e9_printf("[br0285] tp[1]: 0x%08X\n", tp[1]);
    e9_printf("[br0285] tp[2]: 0x%08X\n", tp[2]);
    e9_printf("[br0285] tp[NUM_PAGS_VM - 1]: 0x%08X\n", tp[NUM_PAGS_VM - 1]);
}

```

### 3. Directorio de páginas



```
#include "e9_system.h" /* e9_printf */
#include "kernel.h"
#include <sys/vm.h> /* I386_VM_PRESENT, ... , I386_CR0_PG */

typedef unsigned pte_t;
typedef unsigned pde_t;

#define BIOS_DATE 0x00FFFF5
#define CEIL_DIV(x, y) (((x) + (y) - 1) / (y))
#define MEM_SIZE_MB 128
#define MB_TO_BYTE (1024 * 1024)
#define NUM_PAGS_VM (MEM_SIZE_MB * MB_TO_BYTE) / PAGE_SIZE /* Da 32768 bytes */
#define NUM_PAGS_TP CEIL_DIV(NUM_PAGS_VM * sizeof(pte_t), PAGE_SIZE)
#define NUM_PAGS_DP CEIL_DIV(NUM_PAGS_TP * sizeof(pde_t), PAGE_SIZE)

pte_t rm[(NUM_PAGS_TP + NUM_PAGS_DP + 1) * PAGE_SIZE / sizeof(pde_t)];

void setup_paging(void)
{
    int i;
    unsigned marco_tp, m, cr0, marco_dp;
    pde_t* tp;
    pde_t* dp;

    e9_printf("\n***Yzan Martinez Manzano***\n");
    e9_printf("[br0285] Dir: 0x%08X\n", kinfo.data_base);

    e9_printf("[br0285] Valor en dir. fisica. 0x000FFFF5: %s\n", (char*) (BIOS_DATE -
kinfo.data_base));
```

```

e9_printf("[br0285] Tamano de la Memoria %u MB\n", MEM_SIZE_MB);
e9_printf("[br0285] Numero de Paginas de la Memoria Virtual %u\n", NUM_PAGS_VM);
e9_printf("[br0285] Numero de entradas de la Tabla de Paginas %u\n", NUM_PAGS_TP);
e9_printf("[br0285] Numero de entradas del Directorio de Paginas %u\n", NUM_PAGS_DP);

tp = (pde_t*) (((unsigned)rm) + PAGE_SIZE - 1) & ~(PAGE_SIZE - 1));
e9_printf("[br0285] Dir. tp(abs): 0x%08X\n", ((unsigned)tp) + kinfo.data_base);
e9_printf("[br0285] Offset del tp: 0x%08X\n", ((unsigned)tp) + kinfo.data_base &
0xFFFF);

for (i = 0; i < NUM_PAGS_VM; i++) {
    unsigned marco = i;
    tp[i] = (marco << I386_VM_PT_ENT_SHIFT) | I386_VM_PRESENT | I386_VM_WRITE |
I386_VM_USER;
}

e9_printf("[br0285] tp[0]: 0x%08X\n", tp[0]);
e9_printf("[br0285] tp[1]: 0x%08X\n", tp[1]);
e9_printf("[br0285] tp[2]: 0x%08X\n", tp[2]);
e9_printf("[br0285] tp[NUM_PAGS_VM - 1]: 0x%08X\n", tp[NUM_PAGS_VM - 1]);

marco_tp = (((unsigned)tp + kinfo.data_base) >> I386_VM_PT_ENT_SHIFT);
e9_printf("[br0285] Marco de TP: 0x%08X\n", marco_tp);

dp = (pde_t*) (((unsigned)tp) + (NUM_PAGS_VM * sizeof(pte_t)));
e9_printf("[br0285] Dir. dp(abs): 0x%08X\n", ((unsigned)dp) + kinfo.data_base);
e9_printf("[br0285] Offset del dp: 0x%08X\n", ((unsigned)dp) + kinfo.data_base &
0xFFFF);

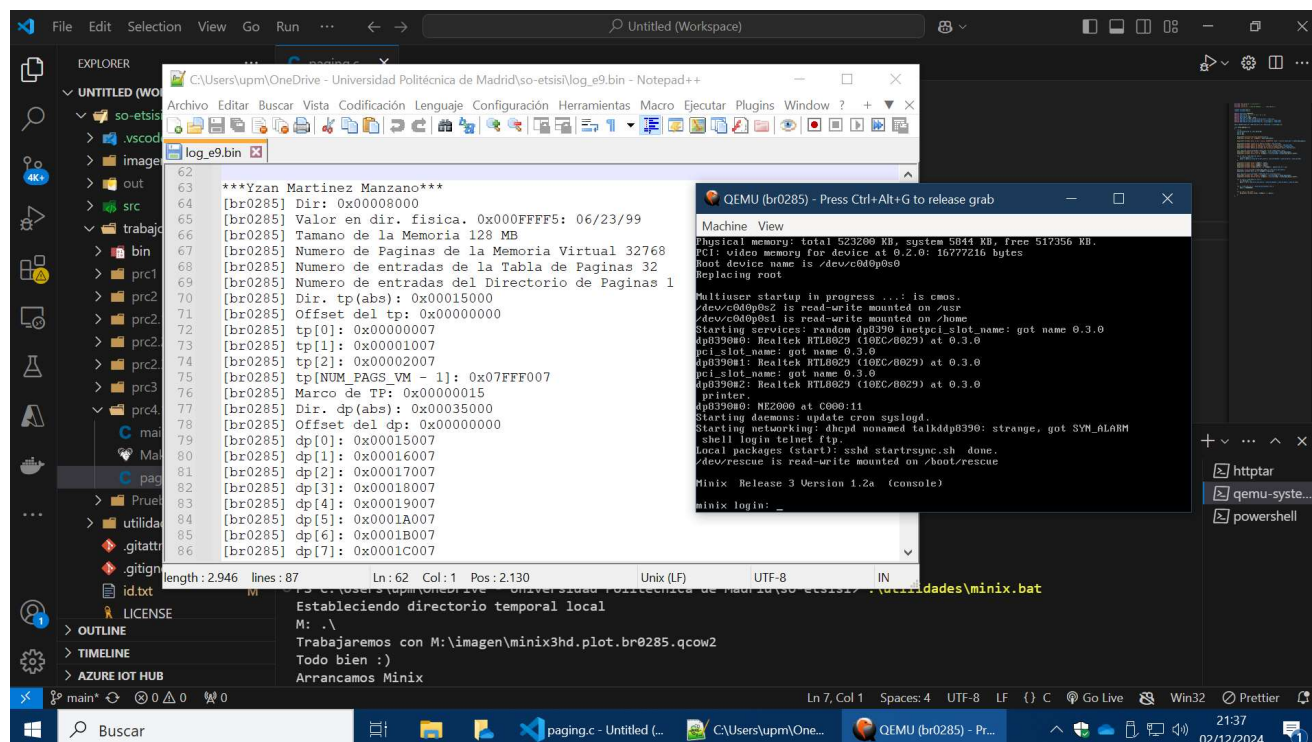
for (i = 0; i < NUM_PAGS_TP; i++) {
    m = marco_tp + i;
    dp[i] = (m << I386_VM_PT_ENT_SHIFT) | I386_VM_PRESENT | I386_VM_WRITE |
I386_VM_USER;
}

for (i = NUM_PAGS_TP; i < I386_VM_DIR_ENTRIES; i++) {
    dp[i] = 0x00000000;
}

for (i = 0; i < 8; i++) {
    e9_printf("[br0285] dp[%d]: 0x%08X\n", i, dp[i]);
}
}

```

## 4. Registros de CR0 y CR3



```
#include "e9_system.h" /* e9_printf */
#include "kernel.h"
#include <sys/vm.h> /* I386_VM_PRESENT, ... , I386_CR0_PG */

typedef unsigned pte_t;
typedef unsigned pde_t;

#define BIOS_DATE 0x00FFFF5
#define CEIL_DIV(x, y) (((x) + (y) - 1) / (y))
#define MEM_SIZE_MB 128
#define MB_TO_BYTE (1024 * 1024)
#define NUM_PAGS_VM (MEM_SIZE_MB * MB_TO_BYTE) / PAGE_SIZE
#define NUM_PAGS_TP CEIL_DIV(NUM_PAGS_VM * sizeof(pte_t), PAGE_SIZE)
#define NUM_PAGS_DP CEIL_DIV(NUM_PAGS_TP * sizeof(pde_t), PAGE_SIZE)

pte_t rm[(NUM_PAGS_TP + NUM_PAGS_DP + 1) * PAGE_SIZE / sizeof(pde_t)];

void setup_paging(void)
{
    int i;
    unsigned marco_tp, m, cr0, marco_dp;
    pde_t* tp;
    pde_t* dp;

    e9_printf("\n***Yzan Martinez Manzano***\n");
    e9_printf("[br0285] Dir: 0x%08X\n", kinfo.data_base);

    e9_printf("[br0285] Valor en dir. fisica. 0x00FFFF5: %s\n", (char*) (BIOS_DATE -
kinfo.data_base));
```

```

e9_printf("[br0285] Tamano de la Memoria %u MB\n", MEM_SIZE_MB);
e9_printf("[br0285] Numero de Paginas de la Memoria Virtual %u\n", NUM_PAGS_VM);
e9_printf("[br0285] Numero de entradas de la Tabla de Paginas %u\n", NUM_PAGS_TP);
e9_printf("[br0285] Numero de entradas del Directorio de Paginas %u\n", NUM_PAGS_DP);

tp = (pde_t*) (((unsigned)rm) + PAGE_SIZE - 1) & ~(PAGE_SIZE - 1));
e9_printf("[br0285] Dir. tp(abs): 0x%08X\n", ((unsigned)tp) + kinfo.data_base);
e9_printf("[br0285] Offset del tp: 0x%08X\n", ((unsigned)tp) + kinfo.data_base &
0xFFFF);

for (i = 0; i < NUM_PAGS_VM; i++) {
    unsigned marco = i;
    tp[i] = (marco << I386_VM_PT_ENT_SHIFT) | I386_VM_PRESENT | I386_VM_WRITE |
I386_VM_USER;
}

e9_printf("[br0285] tp[0]: 0x%08X\n", tp[0]);
e9_printf("[br0285] tp[1]: 0x%08X\n", tp[1]);
e9_printf("[br0285] tp[2]: 0x%08X\n", tp[2]);
e9_printf("[br0285] tp[NUM_PAGS_VM - 1]: 0x%08X\n", tp[NUM_PAGS_VM - 1]);

marco_tp = (((unsigned)tp + kinfo.data_base) >> I386_VM_PT_ENT_SHIFT);
e9_printf("[br0285] Marco de TP: 0x%08X\n", marco_tp);

dp = (pde_t*) (((unsigned)tp) + (NUM_PAGS_VM * sizeof(pte_t)));
e9_printf("[br0285] Dir. dp(abs): 0x%08X\n", ((unsigned)dp) + kinfo.data_base);
e9_printf("[br0285] Offset del dp: 0x%08X\n", ((unsigned)dp) + kinfo.data_base &
0xFFFF);

for (i = 0; i < NUM_PAGS_TP; i++) {
    m = marco_tp + i;
    dp[i] = (m << I386_VM_PT_ENT_SHIFT) | I386_VM_PRESENT | I386_VM_WRITE |
I386_VM_USER;
}

for (i = NUM_PAGS_TP; i < I386_VM_DIR_ENTRIES; i++) {
    dp[i] = 0x00000000;
}

for (i = 0; i < 8; i++) {
    e9_printf("[br0285] dp[%d]: 0x%08X\n", i, dp[i]);
}
}

```